	Hence, by first law of thermodynamics, internal energy increases.	
5a	As current passes through the filament, mobile charge carriers (electrons) transfer kinetic energy to the lattice ions of the filament causing the filament to heat up gradually Filament produces light when it is at a sufficiently high temperature Heating process takes time and hence, full brightness is not immediately achieved.	B1 B1
5b	By potential divider principle, $V_{16\Omega} = \frac{16}{16+14} \times 12$ $= 6.4$ V	C1 A1
5ci	$V_{\text{one lamp}} = 6.0$ V Potential difference across XY = 6.4 – 6.0 = 0.4 V	 A1

5cii	Point Y is at a higher potential.	A1
5d	Current through resistors = $\frac{12}{14+16}$ $= 0.40$ A Current through lamps = 0.50 – 0.40 = 0.10 A	M1 A0
5e	$\frac{\text{total power dissipated by the lamps}}{\text{total power produced by battery}}$ $= \frac{VI_{lamps}}{VI_{battery}}$ $= \frac{12 \times 0.10}{12 \times 0.50}$ $= 0.20$	 C1 A1
5f	No change to potential difference across lamps and hence (by $\frac{V^2}{R}$), power of lamps unchanged (Since effective resistance of circuit increases,) total current decreases and so power produced by battery decreases Ratio increases.	M1 A1
6a		